

IV. POP3

AIM:

To write a java socket program to implement a simple POP3

ALGORITHM:

1. Import all necessary packages
2. Create a string pop3server and with the "pop.gmail.com" string
3. Declare user mail and password as strings
4. Create a socket and bufferedreader
5. Retrieve the mail details of the user
6. Retrieve the email count in the user's account
7. Display the contents of the email
8. Stop

PROGRAM:

```
import java.io.*;
import java.net.*;
import javax.net.ssl.SSLSocketFactory;

public class pop3{
    public static void main(String[] args) {
        String pop3Server = "pop.gmail.com";
        int pop3Port = 995;
        String user = "abc@gmail.com";
        String pass = "12345678";

        try (
            // Create a secure socket connection to the POP3 server
            Socket socket = SSLSocketFactory.getDefault().createSocket(pop3Server, pop3Port);
            BufferedReader in = new BufferedReader(new
InputStreamReader(socket.getInputStream()));
            PrintWriter out = new PrintWriter(new OutputStreamWriter(socket.getOutputStream()),
true)
        ) {
            // Read the initial response from the server
            String line;
            while ((line = in.readLine()) != null) {
                System.out.println(line);
                if (line.startsWith("+OK")) {
                    break;
                }
            }

            // Send USER command
            out.println("USER " + user);
            line = in.readLine();
            System.out.println(line);
        }
    }
}
```

```

// Send PASS command
out.println("PASS " + pass);
line = in.readLine();
System.out.println(line);

// Send STAT command to get the number of messages
out.println("STAT");
line = in.readLine();
System.out.println(line);

if (line.startsWith("+OK")) {
    int msgCount = Integer.parseInt(line.split(" ")[1]);
    for (int i = 1; i <= msgCount; i++) {
        // Retrieve each message
        out.println("RETR " + i);
        boolean msgStarted = false;
        while ((line = in.readLine()) != null) {
            if (line.equals(".")) {
                break;
            }
            if (msgStarted) {
                System.out.println(line);
            }
            if (line.startsWith("+OK")) {
                msgStarted = true;
            }
        }
    }
}

// Send QUIT command to close the connection
out.println("QUIT");
in.readLine(); // Read the response for QUIT command
} catch (IOException e) {
    e.printStackTrace();
}
}
}

```

SAMPLE INPUT AND OUTPUT:

```
GA Command Prompt
--3D"max-width: 672px; min-width: 154px; border-collapse: separate; border-s=
pacing: 0 0.2em;" width=3D"100%" role=3D"presentation"><tbody><tr height=3D=
"24px"><td></td></tr><tr><td><a href=3D"https://docs.google.com/forms?usp=
=3Dmail_form_link" style=3D"color: #424242; font-size: 13px;">Create your o=
wn Google Form</a></td></tr></tbody></table></div></body></html>
--00000000000d1fa3d05ed1899f0--

C:\Users\admin\Documents\cs2249\ex 4\4d>java pop3
+OK Gpop ready for requests from 111.93.231.194 w9mb46284332jal
+OK send PASS
+OK Welcome.
+OK 379 259289857
MIIME-Version: 1.0
x-no-auto-attachment: 1
Received: by 2002:a92:c24f:0:0:0:0:0; Wed, 26 Oct 2022 09:22:43 -0700 (PDT)
Date: Wed, 26 Oct 2022 09:22:43 -0700
Message-ID: <CAEVaNrXMKK3Lm9Hs4a-2rLjtC8yhMHfg3SwOyUpDv2-dBK0G4Q@mail.gmail.com>
Subject: Get the official Gmail app
From: Gmail Team <mail-noreply@google.com>
To: HARISHBALA P CSE <2022cs0016@svce.ac.in>
Content-Type: multipart/alternative; boundary="000000000002ec65205ebf26f0a"

--000000000002ec65205ebf26f0a
Content-Type: text/plain; charset="UTF-8"
Content-Transfer-Encoding: quoted-printable

Get the official Gmail app
[image: Gmail]

Get the official Gmail app
The best features of Sri Venkateswara College of Engineering Mail are only
available on your phone and tablet with the official Gmail app. Download
the app or go to gmail.com on your computer or mobile device to get started=
..
```

RESULT:

Thus, a java socket program to demonstrate POP3 is implemented.